

Academics

2017-2022	B.Tech + M.Tech (Dual Degree) in Computer Science and Engineering, IIT Kharagpur
CGPA	8.60/10.0 (Ongoing)

Research Interest

Heuristic Search | Natural Language | Visual Reasoning | Neuro-Symbolic AI | Deep Learning | Reinforcement Learning

Selected Projects

- Heuristic Search Algorithms for Complex Combinatorial** Sept 2019 - Present
Guide : Prof . Partha Pratim Chakrabarti, Dept. of Computer Science and Engineering, IIT Kharagpur
- Designed a multi-objective heuristic search algorithm for the Fair Market Allocation Problem.
 - Designed a heuristic search algorithm that reduces search tree given the knowledge of "Island" nodes.
- Multimodal Deep Learning** April 2020 - July 2020
Guide : Prof . Humphrey Shi, Department of Computer & Information Science, University of Oregon
- Innovated an End-to-End Transformer based Image Captioning and Visual Question Answering system.
 - Conceptualized an unsupervised object representation learning method capable of capturing semantic relationships.
- Language Driven Outfit Style Enhancement** Jan 2020 - April 2020
Guide : Prof . Abir Das, Dept. of Computer Science and Engineering, IIT Kharagpur
- Innovated a tool using recent developments in generative models for styling one's clothing with textual description.
 - StackGAN was used to generate texture from textual description, followed by Mask RCNN segmenting the clothing region which was then overlaid with the desired texture using TextureGAN.
- Motion Planner for UGV** Oct 2019 - Jan 2020
Guide : Prof . Debashish Chakravarty, Dept. of Mining Engineering, IIT Kharagpur
- Developed an integrated local planner and controller system based on reinforcement learning techniques using LIDAR, encoder and GPS sensor data. Gym-gazebo simulator integrated with Husky UGV was used for training purpose.
- Hybrid A* Path Planner** Sept 2018 - July 2019
Guide : Prof . Debashish Chakravarty, Dept. of Mining Engineering, IIT Kharagpur
- Implemented a search-based motion planner for non-holonomic vehicles navigating in unconstrained environments.
 - Integrated the planner with Gazebo ROS framework and deployed successfully on AGV, IIT KGP's autonomous car.
- Eklavya 6.0** May 2019 - June 2019
Guide : Prof . Debashish Chakravarty, Dept. of Mining Engineering, IIT Kharagpur
- Developed an autonomous navigation system with Hybrid-A* path planner and a computer vision pipeline for lane detection, obstacle avoidance and waypoint generation. (Published)
 - Eklavya 6.0 achieved 2nd position in Auto-Nav Challenge at 27th IGVC held at Oakland University in Rochester, Michigan.

Publications

- Real-time Lane Detection, Fitting and Navigation for Unstructured Environments** in *International Conference of Image, Video Processing and Artificial Intelligence, China(2019)* : S Shrivastava, et al.
- A Prototype of an Intelligent Ground Vehicle for Constrained Environment: Design and Development** in *IEEE International Conference on Control and Robots, South Korea(2019)* : S Shrivastava, et al.

Technical Skills

Languages	C/C++, Python, Prolog, Java
Libraries & Tools	Pytorch, TensorFlow, CUDA, ROS, Open CV, OMPL, Open MP, Open MPI, Gazebo
Others	Git, CMake, Maven

Coursework

University	Artificial Intelligence Natural Language Processing Information Retrieval Machine Learning Deep Learning Programming and Data Structures Algorithms Software Engineering Computer Organization and Architecture Compilers Operating Systems Computer Networks Probability and Statistics Linear Algebra High Performance Parallel Programming
Online	16-782 Planning and Decision-making in Robotics (CMU) Reinforcement Learning(Deepmind)

Achievements

- Secured All India Rank **319** amongst **0.2 million** candidates in **Joint Entrance Examination – Advanced 2017**.
- Recipient of **Kishore Vaigyanik Protsahan Yojana** scholarship awarded by **Government of India**.
- Letter of appreciation by the **Ministry of Human Resource and Development, Government of India** in class 10.